

Site Name -

Pre Commissioning Checklist for Tower Works

1st Stage Inspection – At the time of PCC / Post PCC

SN	Check Points	Observations
1	Date of Inspection	
2	Foundation Drawing Number	
3	Excavation Line – Check whether excavation has been done till 500 mm all around foundation	
4	Depth of foundation	As per drawing – Actual -
5	PCC thickness	
6	Cement – Make, Grade, Month and Year of Manufacture	
7	Check that PCC is done with at least M10 grade concreting (Nominal Mix – 50 Kg Cement, 160 Kg Fine Aggregate, 320 Kg Coarse Aggregate)	
8	Curing of PCC	
9	Back filling with layers for proper water compaction	

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2nd Stage Inspection – Reinforcement Checking

SN	Check Points	Observations
1	Centre Line Alignment of Footings / Pedestals (As per drawing)	
2	Reinforcement make, test certificate for steel from approved laboratory	
3	Bar cutting and bending (As per BBS drawing)	
4	Site in Dry condition / Water logged area	
5	Reinforcement placement as per drawing in Main Footing	
6	Reinforcement placement as per drawing in Footing Strip	
7	Reinforcement placement as per drawing in Pedestals	
8	Pedestal Formwork / shuttering plumb	
9	Chairs Adequacy	
10	Cover block Adequacy	
11	Mode of Concreting planned (RMC / Weight Batcher)	
12	Availability of Material for casting (if not RMC)	
13	Availability of water for casting (if not RMC)	
14	Any other comments -	
15	Date of Inspection	

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3rd Stage Inspection – Casting of Footing and 1st Lift of Pedestal

SN	Check Points	Observations
1	Cement – Make, Grade, Month and Year of Manufacture	
2	Cement – Quantity Adequate	
3	Coarse / Fine Aggregates – Quantity / Quality Adequate	
4	Shuttering Quality and Alignment	
5	Mix Proportion adopted (as per design mix)	
6	Mode of Concreting (RMC / Weight Batchers)	
7	Water cement Ratio & Vibrator	
8	Slump Test – At Start	
9	Slump Test – At mid point	
10	Slump Test – Before closing	
11	Cube filling (as per clause 15 of IS 456)	
12	Centre Line of Pedestal and Plum Check of Pedestal Shuttering	
13	Curing of Pedestal	
14	Any other comments -	
15	Date of Inspection	

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3rd Stage Inspection – Casting of Footing and 2nd Lift of Pedestal

SN	Check Points	Observations
1	Cement – Make, Grade, Month and Year of Manufacture	
2	Cement – Quantity Adequate	
3	Coarse / Fine Aggregates – Quantity / Quality Adequate	
4	Shuttering Quality and Alignment	
5	Mix Proportion adopted (as per design mix)	
6	Mode of Concreting (RMC / Weight Batchers)	
7	Water cement Ratio & Vibrator (as per design mix)	
8	Slump Test – At Start	
9	Slump Test – At mid point	
10	Slump Test – Before closing	
11	Cube filling (as per clause 15 of IS 456)	
12	Curing of pedestal	
13	Any other comments -	
14	Date of Inspection	

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4th Stage Inspection – CIP Fixing and Casting of 3rd Lift of Pedestal

SN	Check Points	Observations
1	CIP level with respect to Top of Pedestal	
2	Water level check between CIP (if some portion of tower is in water logged area)	
3	CIP Plan placement and bolt tightness	
4	Top of Pedestal with respect to NGL in mm	
5	Plum check of Pedestal shuttering – Pre-casting	
6	Cement – Make, Grade, Month and Year of Manufacture	
7	Cement – Quantity Adequate	
8	Coarse / Fine Aggregates – Quantity / Quality Adequate	
9	Mix Proportion adopted (as per design mix)	
10	Mode of Concreting (RMC / Weight Batchter)	
11	Water cement Ratio & Vibrator (as per design mix)	
12	Slump Test – At Start	
13	Slump Test – At mid point	
14	Plumb check of Pedestal shuttering – Post casting	
15	Cube filling (as per clause 15 of IS 456)	
16	Curing of pedestal	
17	Any other comments -	
18	Date of Inspection -	

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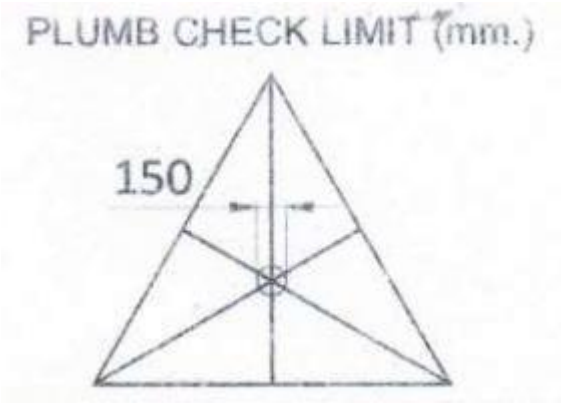
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5th Stage Inspection – Tower Post Completion check

SN	Check Points	Observations
1	Availability of all accessories as per Drawing	
2	GI strip connected	
3	Painting Done	
4	Random Tightness check	
5	Vertically (use of Plumb) – (Refer sketch below)	
	PLUMB CHECK LIMIT (mm) 	
6	Overall comments (if any)	
7	Any other comments	
8	Date of Inspection	

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INSPECTION CHECK LIST FOR FOUNDATION OF TOWERS

The Tower foundation will be inspected at each stage by the inspections from WC Railways and firm as per the following inspection plan. Clearance will be given only after these tests are passed.

SN	Description of Items to be checked	OK / Not OK	Remarks
I	Excavation of Pit. a) Correct marking of pit location with reference to bench marks. b) Checking of alignment and measurement of the pit before PCC		
II	Quality of materials		
a)	CEMENT: standard make (OPC- 43 grade) ISI marked		
b)	FINE AGGREGATE (Sand): Be clean & shall not contain clay or any deleterious material		
c)	Coarse aggregate (Jelly/ Kankar of 20 mm / 40mm) (a) flaky or elongated (b) nearly cubic shape		
d)	Steel Rods used for Reinforcement: As per IS 1786, Fe415/Fe/500D		
e)	Water used shall be clean and drinking quality and shall not be salty.		
III	RCC Stages		
a)	Formwork Checking (a) Formwork is properly aligned (b) Check if the formwork is water- tight (c) Supports are properly placed and fixed (d) Check that width, depth and height of form work are correct, as per design.		
b)	Reinforcement Checking (a) Steel rods are placed as per drawing (b) Check reinforcement bars are straight (c) Dia of reinforcement bars is as per design (d) Spacing of reinforcement bar is as per design (e) Hooks & bends are placed as specified by structural designer (f) Lap length is adequate (g) Check the clear cover (h) Cover blocks are enough (i) The rods are tied properly & tightly with binding wire (j) Check quantity (count) as per drawing.		
c)	Pre-concreting Checking (a) Check all required materials and tools are available at site (b) Check that adequate lighting is available (if it is planned to work at night)		

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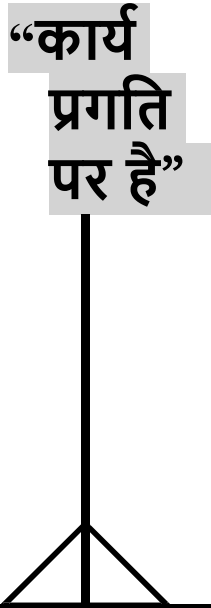
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SN	Description of Items to be checked	OK / Not OK	Remarks
	(c) Check the form work supports are intact and not displaced.		
d)	Checking during concreting (a) Check slump for every 20 cu. m of concrete for proper water and cement ratio. (b) Concrete cubes are made for testing (Concrete shall be collected from 3 or 4 random mixes for making concrete cubes (150 mm) which would be sent for third party examination or strength testing at site (c) Check that vibrator is used adequately and properly (d) Check that concrete surface is level (e) Check the finishing of concrete surface is level (f) Check that adequate curing is done		
IV	Check for escarpment: There is no such local site condition which necessitate change in design of concrete surface of foundation		
V	Check that barricading and luminous strips have been provided surrounding tower pits		
VI	Display board for site as per diagram is placed – 		
VII	Curing of concreting		
VIII	Back filling with layers for proper water compaction		

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INSPECTION CHECK LIST FOR ERECTION OF TOWERS

The Tower foundation will be inspected at each stage by the inspections from WC Railways and firm as per the following inspection plan. Clearance will be given only after these tests are passed.

SN	Description of Items to be checked	OK / Not OK	Remarks
I	Quality of Raw materials: Check that the raw material has been duly inspected by RITES (attach RITES Inspection Certificate)		
II	Inspection of Fabricated parts: (a) Visual examination and quantity verification will be carried out for the offered lot. (b) No rough edges shall be permitted in the entire structure. (c) Check that fabricated tower has been duly inspected by RITES (attach RITES Inspection Certificate)		
III	Erection of Tower: a) Safety precautions for men & materials are in place. b) Vertically check for each segment and tower. c) Uniform fixing all members. d) Fixing of ladders, safety rings, antenna, lightning arrestors, aviation lamps, earthing and fencing on the tower portion.		
IV	Check for escarpment: There is no such local site condition which necessitate change in design of tower of foundation.		
V	Painting (after erection): 1 coat yellow primer & 2 coats white & orange bands (as per standard practice)		

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